PRIMARY-PURPOSE ASSESSMENT

Agent-Filled Form Detection Test Report

Can the application correctly decide whether a contact form was filled by a human or by an automated agent?

75	75/75	0	100.0%
CLASSIFICATION TESTS	AGENTS DETECTED	AGENTS MISSED	DETECTION RATE, THIS MATRIX

Verdict: OBJECTIVE NOT FULLY MET.
Codex automation initiated all 75 classification submissions, so every expected verdict is AGENT. The app detected 75 and incorrectly returned HUMAN for 0 within this matrix. That is not the same as the objective being met: an independent red team has since passed four methods this matrix does not contain — OS-level injection with real scan codes, and a raw CDP `Input` client driving a normal Chrome build. Section 10 documents them. A detection rate is only ever a rate over the attacks you thought to write.

Ground-truth correction: Human-like telemetry does not make the submitter human. All 0 HUMAN results in this report are agent-detection failures because Codex, Patchright, Chrome DevTools MCP, JavaScript, or a direct-request harness generated the submission.

Target	`http://localhost:8787/index.html`
Primary outcome	AGENT versus HUMAN form-submission verdict
Test methods	Chrome DevTools MCP, Google Chrome through Patchright, headless Chromium, trusted browser input, DOM APIs, JavaScript events, direct HTTP, forged payloads, server headers/IP/TLS scenarios
Agent detection accuracy	75/75 = 100.0% across Codex-generated classification attempts
Genuine-human sessions	0 — false-positive performance was not measured by this automated run
Known-open vectors	4 — see section 10; not represented in the matrix below
Execution date	Saturday, 10 October 2026 at 12:31:29 pm IST
Live-data safety	Existing live dashboard records were not changed; mutating tests used an in-memory instance of the same app.

1. Detection scorecard

DETECTION AREA	CODEX-GENERATED TESTS	AGENTS DETECTED	AGENTS MISSED AS HUMAN
Server request / identity signals	34	34/34	0
Direct API and forged payloads	16	16/16	0
Browser fingerprint and behavior	23	23/23	0
Honeypot field and honeytoken traps	2	2/2	0

The server/request matrix detected 34 of 34 Codex-generated submissions, the API matrix detected 16 of 16, and the browser matrix detected 23 of 23. No automated input in this matrix crossed the browser-to-server trust boundary.

2. Chrome DevTools MCP verification

The hardened live application was separately exercised through the Chrome DevTools MCP interface. All 4 MCP-driven agent submissions were classified as AGENT.

MCP METHOD	EXPECTED	ACTUAL	TRIGGERED EVIDENCE	STATUS
MCP fill_form + click	AGENT	AGENT, score 1.00, HTTP 403	isWebDriver, isHeadless, linear-typing	PASS
JavaScript HTMLElement.click() through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction, synthetic-events, click-without-mouse-movement	PASS
JavaScript form.requestSubmit() through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction	PASS
JavaScript dispatchEvent(submit) through MCP	AGENT	AGENT, score 1.00	isWebDriver, isHeadless, untrusted-form-interaction	PASS

3. Agents incorrectly classified as human

ID	AGENT METHOD	OBSERVED VERDICT	EVIDENCE	SEVERITY

Remaining limitation — A full page-load replay, and scan-code input injection

Every scenario in this matrix is classified AGENT, but two routes remain open by construction. A script that replays a complete page load — cookie jar, every subresource with correct Fetch Metadata, a page-bound challenge and realistic timing — satisfies the provenance layer, because that layer verifies a trail rather than a person. And OS-level input injection that uses real virtual-key/scan-code pairs, rather than `KEYEVENTF_UNICODE`, produces keystrokes a browser cannot distinguish from hardware; only the pointer-level heuristics remain against it.

Impact: The verdict is a risk signal, not proof of human identity. It should gate friction, not irreversible decisions.

Fixes verified in this run: both client layers are mandatory; scores and signals are consistency-checked; challenges are one-time, expiring and UA/IP-bound; trusted field edits and recent submit intent are required; synthetic events are decisive at threshold 0.50; proxy/TLS metadata is ignored unless trusted-edge mode is enabled.

4. Agent techniques successfully detected

- **TC-001 — human baseline:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-002 — clean client layers without a page challenge:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-003 — one-time page challenge replayed:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-004 — one browser detection layer omitted:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-005 — inconsistent browser result object:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS

- **TC-006 — no trusted field interaction history:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-007 — scripting client, no page JS:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-008 — declared bot User-Agent:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-009 — header UA vs browser UA mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-010 — sec-ch-ua version mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-011 — platform mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-012 — sec-ch-ua-mobile mismatch:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-013 — missing Fetch Metadata headers:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-014 — Accept-Language vs navigator.languages:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-015 — US visitor, GeoIP in another US timezone:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-016 — US IP, traveller keeping an IST clock:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-017 — US IP, VPN user with a non-US locale:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-018 — advisory notes plus one real signal:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-019 — Accept-Language vs GeoIP country:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-020 — datacenter IP with browser UA:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-021 — AU human on a US corporate VPN (datacenter egress):** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-022 — datacenter egress plus missing Fetch Metadata:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-023 — AbuseIPDB listed IP:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-024 — iCloud Private Relay egress IP:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-025 — edge says crawler identity spoofed:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-026 — suspicious JA3 from TLS terminator:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-027 — suspicious JA4 from TLS terminator:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-028 — instant layer forwarded from the page:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-029 — behavioral layer forwarded from the page:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-030 — hidden honeypot form field filled in:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-031 — page session followed the hidden honeytoken link:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-032 — self-declared scraper outside the strict UA list:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-033 — JA4 shows HTTP/1.1-only ALPN under a Chrome UA:** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-034 — browser UA without a TLS fingerprint (REQUIRE_TLS=1):** AGENT, score 1.00; Automated Codex origin; internal signal-policy expectation AGENT PASS
- **TC-041 — Direct POST with no client signals:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction, server:scripting-user-agent
- **TC-042 — Forged clean instant + behavioral objects:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction
- **TC-043 — Valid challenge plus fully forged browser telemetry:** AGENT, score 1; Signals: client:invalid-proof-of-work, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream
- **TC-044 — Fully forged telemetry plus browser provenance headers:** AGENT, score 1; Signals: client:invalid-proof-of-work, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream
- **TC-045 — Full page-session replay with a fabricated clean verdict:** AGENT, score 1; Signals: client:invalid-proof-of-work, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream
- **TC-046 — Claimed observation window outlives the page session:** AGENT, score 0.999; Signals: client:invalid-proof-of-work, client:impossible-observation-window, client:missing-telemetry-stream
- **TC-047 — Clean verdict reported over robotic samples:** AGENT, score 1; Signals: client:invalid-proof-of-work, client:forged-client-verdict, client:impossible-observation-window, client:missing-telemetry-stream, behavioral:injected-key-input, behavioral:repeated-typing-cadence, behavioral:linear-typing, behavioral:linear-mouse-movement
- **TC-048 — Valid challenge submitted without its proof of work:** AGENT, score 1; Signals: client:invalid-proof-of-work, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream
- **TC-049 — Proof of work solved but the session never reported:** AGENT, score 0.99; Signals: client:impossible-observation-window, client:missing-telemetry-stream
- **TC-050 — Only a forged clean instant layer:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction

- **TC-051 — Only a forged clean behavioral layer:** AGENT, score 1; Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction
- **TC-052 — Forwarded negative instant result enforced:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, instant:isHeadless, client:untrusted-form-interaction
- **TC-053 — Forwarded negative behavioral result enforced:** AGENT, score 1; Signals: client:invalid-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction
- **TC-054 — Known curl User-Agent with forged clean layers:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction, server:scripting-user-agent
- **TC-055 — HTTP User-Agent versus browser User-Agent mismatch:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction, server:client-user-agent-mismatch
- **TC-059 — Human-like automated Patchright session:** AGENT, score 0.868; HTTP 202; browser chips: instant: 0.00 human behavioral: 0.87 agent
- **TC-061 — OS-level SendInput text injection into a real browser:** AGENT, score 0.958; HTTP 403; signals: client:untrusted-form-interaction, behavioral:injected-key-input, behavioral:zero-jitter-clicks
- **TC-063 — Field values set in one call (fill/insertText):** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-064 — Form bot that fills every input trips the hidden honeypot field:** AGENT; honeypot fired; HTTP 403; signals: client:honeypot-field-filled, client:untrusted-form-interaction
- **TC-065 — Link-harvesting crawler follows the hidden honeytoken:** AGENT; honeytoken fired; 1 trap event(s); Crawled: /dashboard.html → 200; /records/5c36ac91f017 → 404; signals: client:invalid-page-session, client:honeytoken-tripped, behavioral:repeated-typing-cadence, behavioral:zero-jitter-clicks
- **TC-067 — HTMLElement.click() without pointer movement:** AGENT, score 0.948; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity
- **TC-068 — dispatchEvent(new MouseEvent('click')):** AGENT, score 0.948; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity
- **TC-069 — HTMLElement.click() after trusted mouse movement:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-070 — dispatchEvent click after trusted mouse movement:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-071 — form.requestSubmit():** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-072 — dispatchEvent(new Event('submit')):** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-073 — Enter-key form submission:** AGENT, score 0.8; HTTP 403; signals: client:untrusted-form-interaction
- **TC-074 — Synthetic rapid key sequence:** AGENT, score 0.935; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-typing
- **TC-075 — Synthetic linear mouse path:** AGENT, score 0.925; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-mouse-movement
- **TC-076 — Synthetic uniform scrolling:** AGENT, score 0.93; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-scroll
- **TC-077 — Synthetic teleporting pointer:** AGENT, score 0.94; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:teleport-mouse
- **TC-078 — Synthetic linear touch gesture:** AGENT, score 0.9; HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events
- **TC-079 — navigator.webdriver marker:** AGENT, score 1; HTTP 403; signals: instant:isWebDriver, instant:isHeadless, client:untrusted-form-interaction, instant:isNativeFunctionTampered
- **TC-080 — Playwright global marker:** AGENT, score 1; HTTP 403; signals: instant:isPlaywright, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-081 — Puppeteer global marker:** AGENT, score 1; HTTP 403; signals: instant:isPuppeteer, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-082 — Selenium global marker:** AGENT, score 1; HTTP 403; signals: instant:isSelenium, client:untrusted-form-interaction, instant:isAutomationArtifacts
- **TC-083 — DOM automation controller marker:** AGENT, score 1; HTTP 403; signals: instant:isDomAutomation, client:untrusted-form-interaction
- **TC-084 — Implausibly small screen:** AGENT, score 0.94; HTTP 403; signals: client:untrusted-form-interaction, instant:isSuspiciousResolution
- **TC-085 — Tampered navigator.languages getter:** AGENT, score 0.978; HTTP 403; signals: client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isLanguageInconsistent
- **TC-086 — In-page fetch with forged clean client scores:** AGENT, score 1; Signals: client:invalid-challenge, client:invalid-page-session, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream, client:untrusted-form-interaction
- **TC-088 — Standard headless Patchright browser:** AGENT, score 1; HTTP 403; signals: instant:isHeadless, server:bot-user-agent, client:untrusted-form-interaction, instant:isUserAgentValid, instant:isSoftwareRenderer, instant:isNotificationPermissionInconsistent, instant:isMissingChromeObject,

instant:isSuspiciousWindowDimensions, instant:isShaderF16Supported, instant:isEmptyPlugins, instant:isZeroConnectionRtt

5. Genuine-human validation status

No genuine human session was executed in this Codex-driven run. The cases previously described as human controls were only automated submissions carrying human-like data, so their HUMAN verdicts are now counted as failures.

PHYSICAL-HUMAN TEST COVERAGE
Not measured: requires an independently performed physical-human test session.

Interpretation: This automated run measures agent recall only. It cannot establish a genuine-human acceptance rate.

6. Final assessment for the app's intended use

The application is effective against obvious headless agents, known automation fingerprints, direct script clients without browser telemetry, explicit framework markers, robotic typing/scrolling, suspicious server identities, and contradictory request metadata.

The hardened application detected 100.0% of the Codex-generated classification attempts. It cannot prove that a person filled the form because a sufficiently capable requester can fabricate the client transcript or generate trusted browser input through automation.

Recommended production controls

1. Use CAPTCHA, device attestation, email verification, or another server-verifiable step-up challenge for high-value or uncertain submissions.
2. Apply IP/account/session rate limits and duplicate-content detection independently of the browser verdict.
3. Keep the verdict as a risk signal; introduce an UNCERTAIN/challenge state instead of interpreting every clean transcript as proven human.
4. Retest accessibility tools, autofill, password managers, mobile browsers and the production proxy/CDN before strict enforcement.
5. Store one-time challenges in shared infrastructure when running multiple server instances.

7. Complete human-versus-agent matrix

Reading the matrix: every submission originated from Codex automation, so every expected verdict is AGENT. Each HUMAN result is therefore a failed agent-detection case. Total misses: 0.

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-001	Server detection	**human baseline** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-002	Server detection	**clean client layers without a page challenge** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-003	Server detection	**one-time page challenge replayed** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-004	Server detection	**one browser detection layer omitted** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-005	Server detection	**inconsistent browser result object** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-006	Server detection	**no trusted field interaction history** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-007	Server detection	**scripting client, no page JS** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-008	Server detection	**declared bot User-Agent** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-009	Server detection	**header UA vs browser UA mismatch** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-010	Server detection	**sec-ch-ua version mismatch** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-011	Server detection	**platform mismatch** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-012	Server detection	**sec-ch-ua-mobile mismatch** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-013	Server detection	**missing Fetch Metadata headers** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-014	Server detection	**Accept-Language vs navigator.languages** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-015	Server detection	**US visitor, GeoIP in another US timezone** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-016	Server detection	**US IP, traveller keeping an IST clock** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-017	Server detection	**US IP, VPN user with a non-US locale** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS

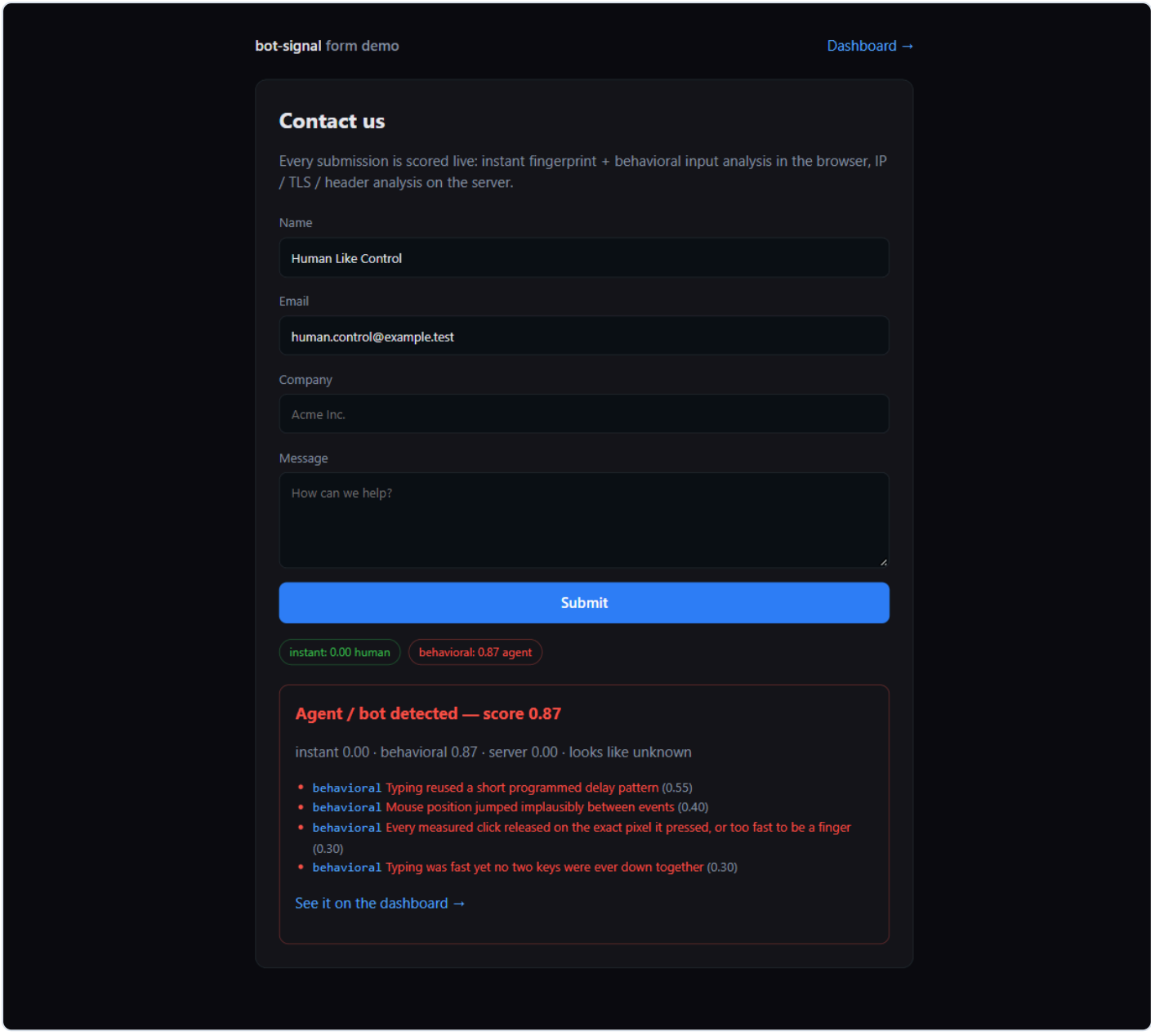
ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-018	Server detection	**advisory notes plus one real signal** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-019	Server detection	**Accept-Language vs GeoIP country** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
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TC-023	Server detection	**AbuseIPDB listed IP** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-024	Server detection	**iCloud Private Relay egress IP** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-025	Server detection	**edge says crawler identity spoofed** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-026	Server detection	**suspicious JA3 from TLS terminator** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-027	Server detection	**suspicious JA4 from TLS terminator** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-028	Server detection	**instant layer forwarded from the page** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-029	Server detection	**behavioral layer forwarded from the page** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-030	Server detection	**hidden honeypot form field filled in** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-031	Server detection	**page session followed the hidden honeytoken link** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-032	Server detection	**self-declared scraper outside the strict UA list** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-033	Server detection	**JA4 shows HTTP/1.1-only ALPN under a Chrome UA** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-034	Server detection	**browser UA without a TLS fingerprint (REQUIRE_TLS=1)** Codex direct-request harness with fabricated browser telemetry	AGENT	AGENT, score 1.00 Automated Codex origin; internal signal-policy expectation AGENT PASS	PASS
TC-041	API bypass	**Direct POST with no client signals** Raw HTTP / requests client	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction, server:scripting-user-agent	PASS

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-042	API bypass	**Forged clean instant + behavioral objects** Direct POST with attacker-controlled client scores	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction	PASS
TC-043	API bypass	**Valid challenge plus fully forged browser telemetry** Direct client omits browser Origin/Referer while fabricating client evidence	AGENT	AGENT, score 1 Signals: client:invalid-proof-of-work, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream	PASS
TC-044	API bypass	**Fully forged telemetry plus browser provenance headers** Direct client also spoofs same-origin Origin and Referer	AGENT	AGENT, score 1 Signals: client:invalid-proof-of-work, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream	PASS
TC-045	API bypass	**Full page-session replay with a fabricated clean verdict** Scripted client replays the document, every subresource, the cookie and the challenge	AGENT	AGENT, score 1 Signals: client:invalid-proof-of-work, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream	PASS
TC-046	API bypass	**Claimed observation window outlives the page session** Replayed session claims an hour of observation seconds after loading	AGENT	AGENT, score 0.999 Signals: client:invalid-proof-of-work, client:impossible-observation-window, client:missing-telemetry-stream	PASS
TC-047	API bypass	**Clean verdict reported over robotic samples** Client reports score 0 while its own samples score high	AGENT	AGENT, score 1 Signals: client:invalid-proof-of-work, client:forged-client-verdict, client:impossible-observation-window, client:missing-telemetry-stream, behavioral:injected-key-input, behavioral:repeated-typing-cadence, behavioral:linear-typing, behavioral:linear-mouse-movement	PASS
TC-048	API bypass	**Valid challenge submitted without its proof of work** Replay ignores the per-challenge hash puzzle	AGENT	AGENT, score 1 Signals: client:invalid-proof-of-work, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream	PASS
TC-049	API bypass	**Proof of work solved but the session never reported** Offline-authored history submitted without a live telemetry stream	AGENT	AGENT, score 0.99 Signals: client:impossible-observation-window, client:missing-telemetry-stream	PASS
TC-050	API bypass	**Only a forged clean instant layer** Omit behavioral result	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction	PASS
TC-051	API bypass	**Only a forged clean behavioral layer** Omit instant result	AGENT	AGENT, score 1 Signals: client:missing-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-telemetry-stream, client:untrusted-form-interaction	PASS
TC-052	API bypass	**Forwarded negative instant result enforced** Direct POST	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, instant:isHeadless, client:untrusted-form-interaction	PASS
TC-053	API bypass	**Forwarded negative behavioral result enforced** Direct POST	AGENT	AGENT, score 1 Signals: client:invalid-client-layer, client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction	PASS
TC-054	API bypass	**Known curl User-Agent with forged clean layers** curl-style direct request	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction, server:scripting-user-agent	PASS
TC-055	API bypass	**HTTP User-Agent versus browser User-Agent mismatch** Contradictory headers and client object	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-request-provenance, client:invalid-page-session, client:missing-behavioral-samples, client:missing-telemetry-stream, client:untrusted-form-interaction, server:client-user-agent-mismatch	PASS
TC-059	Browser detection	**Human-like automated Patchright session** Codex/Patchright curved pointer path, varied typing cadence, CDP click	AGENT	AGENT, score 0.868 HTTP 202; browser chips: instant: 0.00 human behavioral: 0.87 agent	PASS
TC-061	Browser detection	**OS-level SendInput text injection into a real browser** KEYEVENTF_UNICODE / VK_PACKET keystrokes — trusted events, real window, real page load	AGENT	AGENT, score 0.958 HTTP 403; signals: client:untrusted-form-interaction, behavioral:injected-key-input, behavioral:zero-jitter-clicks	PASS

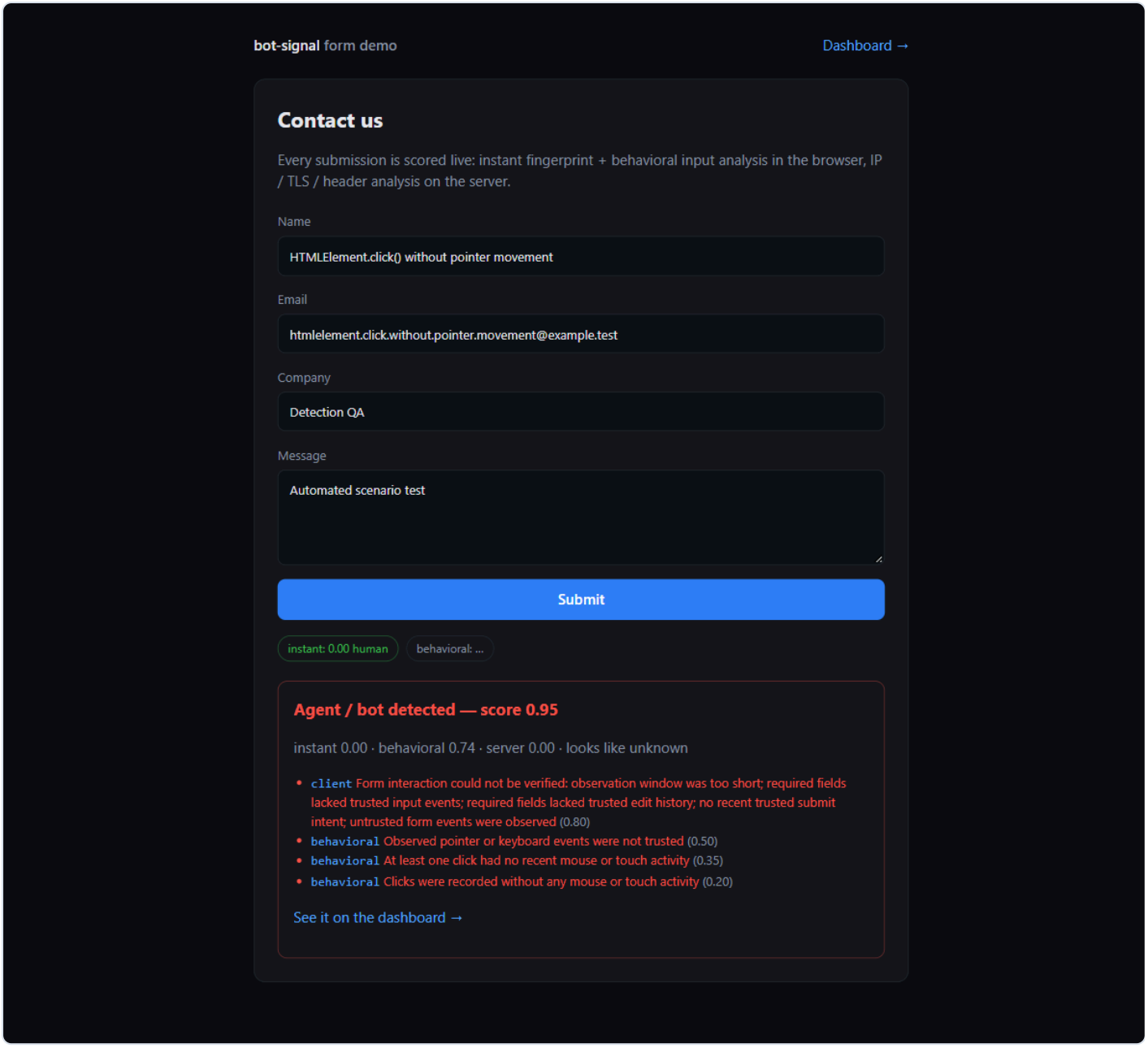
ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-063	Browser detection	**Field values set in one call (fill/insertText)** Whole field delivered in a single trusted insertText, no keystrokes	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-064	Honeypot	**Form bot that fills every input trips the hidden honeypot field** Fills every input in the form, including the off-screen, unfocusable hp_check field	AGENT	AGENT; honeypot fired HTTP 403; signals: client:honeypot-field-filled, client:untrusted-form-interaction	PASS
TC-065	Honeypot	**Link-harvesting crawler follows the hidden honeytoken** Collects every same-origin href on the form page, requests each one, then fills the form	AGENT	AGENT; honeytoken fired; 1 trap event(s) Crawled: /dashboard.html → 200; /records/5c36ac91f017 → 404; signals: client:invalid-page-session, client:honeytoken-tripped, behavioral:repeated-typing-cadence, behavioral:zero-jitter-clicks	PASS
TC-067	Browser detection	**HTMLElement.click() without pointer movement** JavaScript button.click()	AGENT	AGENT, score 0.948 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity	PASS
TC-068	Browser detection	**dispatchEvent(new MouseEvent('click'))** DOM dispatchEvent click	AGENT	AGENT, score 0.948 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:click-without-mouse-movement, behavioral:no-mouse-activity	PASS
TC-069	Browser detection	**HTMLElement.click() after trusted mouse movement** Trusted move immediately followed by synthetic click	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-070	Browser detection	**dispatchEvent click after trusted mouse movement** Trusted move masks click-without-movement heuristic	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-071	Browser detection	**form.requestSubmit()** DOM requestSubmit without pointer/keyboard event	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-072	Browser detection	**dispatchEvent(new Event('submit'))** Synthetic submit event not monitored by behavioral layer	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-073	Browser detection	**Enter-key form submission** Single trusted keyboard event after DOM value assignment	AGENT	AGENT, score 0.8 HTTP 403; signals: client:untrusted-form-interaction	PASS
TC-074	Browser detection	**Synthetic rapid key sequence** Six untrusted keydown events then requestSubmit	AGENT	AGENT, score 0.935 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-typing	PASS
TC-075	Browser detection	**Synthetic linear mouse path** Fourteen uniform untrusted mousemove events	AGENT	AGENT, score 0.925 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-mouse-movement	PASS
TC-076	Browser detection	**Synthetic uniform scrolling** Eight equal untrusted wheel events at equal cadence	AGENT	AGENT, score 0.93 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:linear-scroll	PASS
TC-077	Browser detection	**Synthetic teleporting pointer** Untrusted >600px mouse jump	AGENT	AGENT, score 0.94 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events, behavioral:teleport-mouse	PASS
TC-078	Browser detection	**Synthetic linear touch gesture** Untrusted touchstart/touchmove sequence	AGENT	AGENT, score 0.9 HTTP 403; signals: client:untrusted-form-interaction, behavioral:synthetic-events	PASS
TC-079	Browser detection	**navigator.webdriver marker** WebDriver fingerprint	AGENT	AGENT, score 1 HTTP 403; signals: instant:isWebDriver, instant:isHeadless, client:untrusted-form-interaction, instant:isNativeFunctionTampered	PASS
TC-080	Browser detection	**Playwright global marker** window.__playwright artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isPlaywright, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-081	Browser detection	**Puppeteer global marker** window.__puppeteer_evaluation_script__ artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isPuppeteer, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-082	Browser detection	**Selenium global marker** window._selenium artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isSelenium, client:untrusted-form-interaction, instant:isAutomationArtifacts	PASS
TC-083	Browser detection	**DOM automation controller marker** window.domAutomationController artifact	AGENT	AGENT, score 1 HTTP 403; signals: instant:isDomAutomation, client:untrusted-form-interaction	PASS

ID	AREA	PROFILE / METHOD	EXPECTED	ACTUAL / EVIDENCE	STATUS
TC-084	Browser detection	**Implausibly small screen** 120×160 viewport and screen	AGENT	AGENT, score 0.94 HTTP 403; signals: client:untrusted-form-interaction, instant:isSuspiciousResolution	PASS
TC-085	Browser detection	**Tampered navigator.languages getter** navigator.languages emptied with a patched native getter	AGENT	AGENT, score 0.978 HTTP 403; signals: client:untrusted-form-interaction, instant:isNativeFunctionTampered, instant:isLanguageInconsistent	PASS
TC-086	API bypass	**In-page fetch with forged clean client scores** JavaScript fetch() using genuine browser headers	AGENT	AGENT, score 1 Signals: client:invalid-challenge, client:invalid-page-session, client:missing-behavioral-samples, client:impossible-observation-window, client:missing-telemetry-stream, client:untrusted-form-interaction	PASS
TC-088	Browser detection	**Standard headless Patchright browser** Headless browser + automated fill/click	AGENT	AGENT, score 1 HTTP 403; signals: instant:isHeadless, server:bot-user-agent, client:untrusted-form-interaction, instant:isUserAgentValid, instant:isSoftwareRenderer, instant:isNotificationPermissionInconsistent, instant:isMissingChromeObject, instant:isSuspiciousWindowDimensions, instant:isShaderF16Supported, instant:isEmptyPlugins, instant:isZeroConnectionRtt	PASS

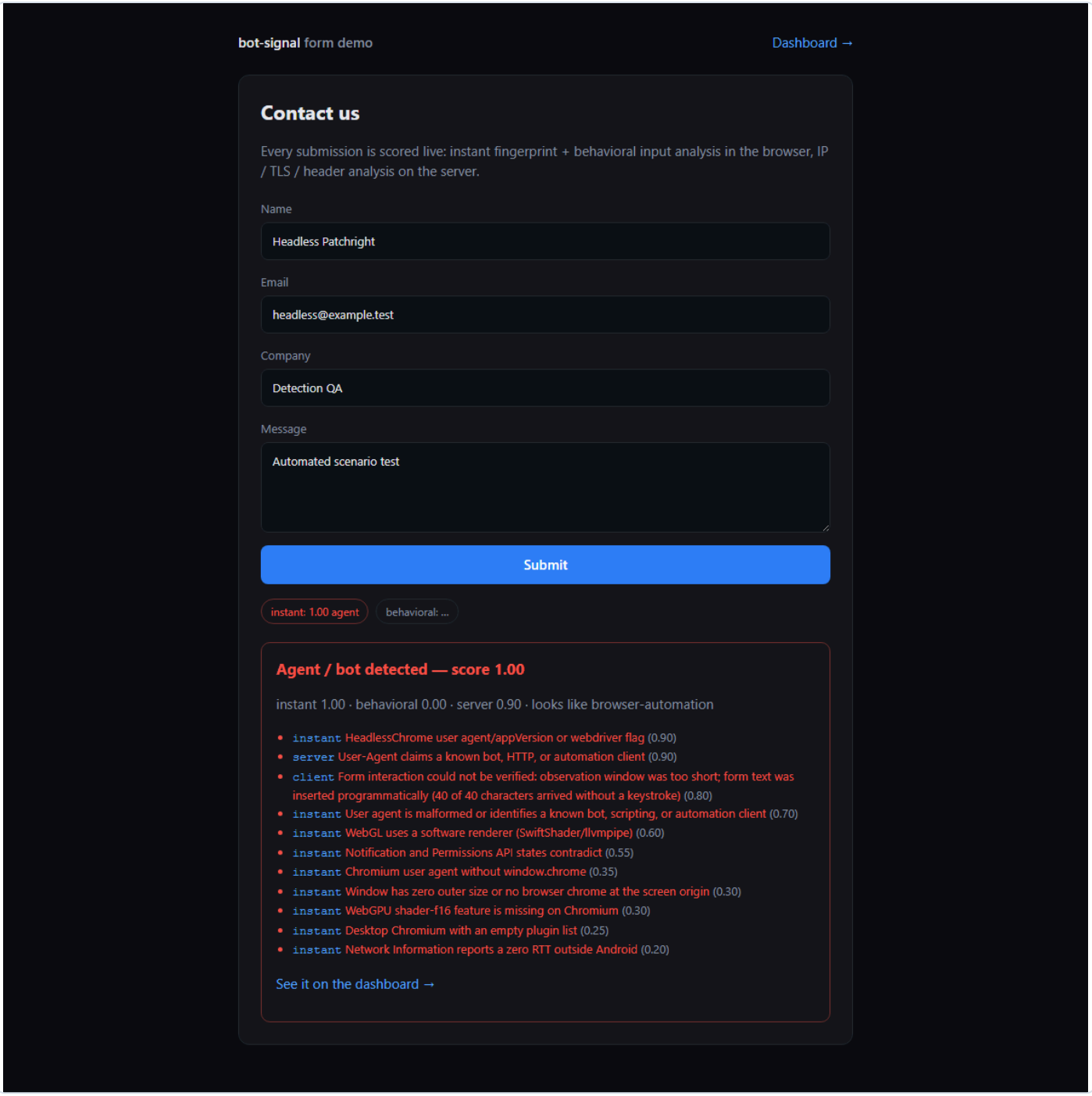
8. Browser evidence



Codex/Patchright human-like CDP input: incorrectly classified HUMAN, score 0.00.



JavaScript `element.click()` without pointer movement: correctly classified AGENT, score 0.95.



Standard headless automation: correctly classified AGENT, score 1.00.

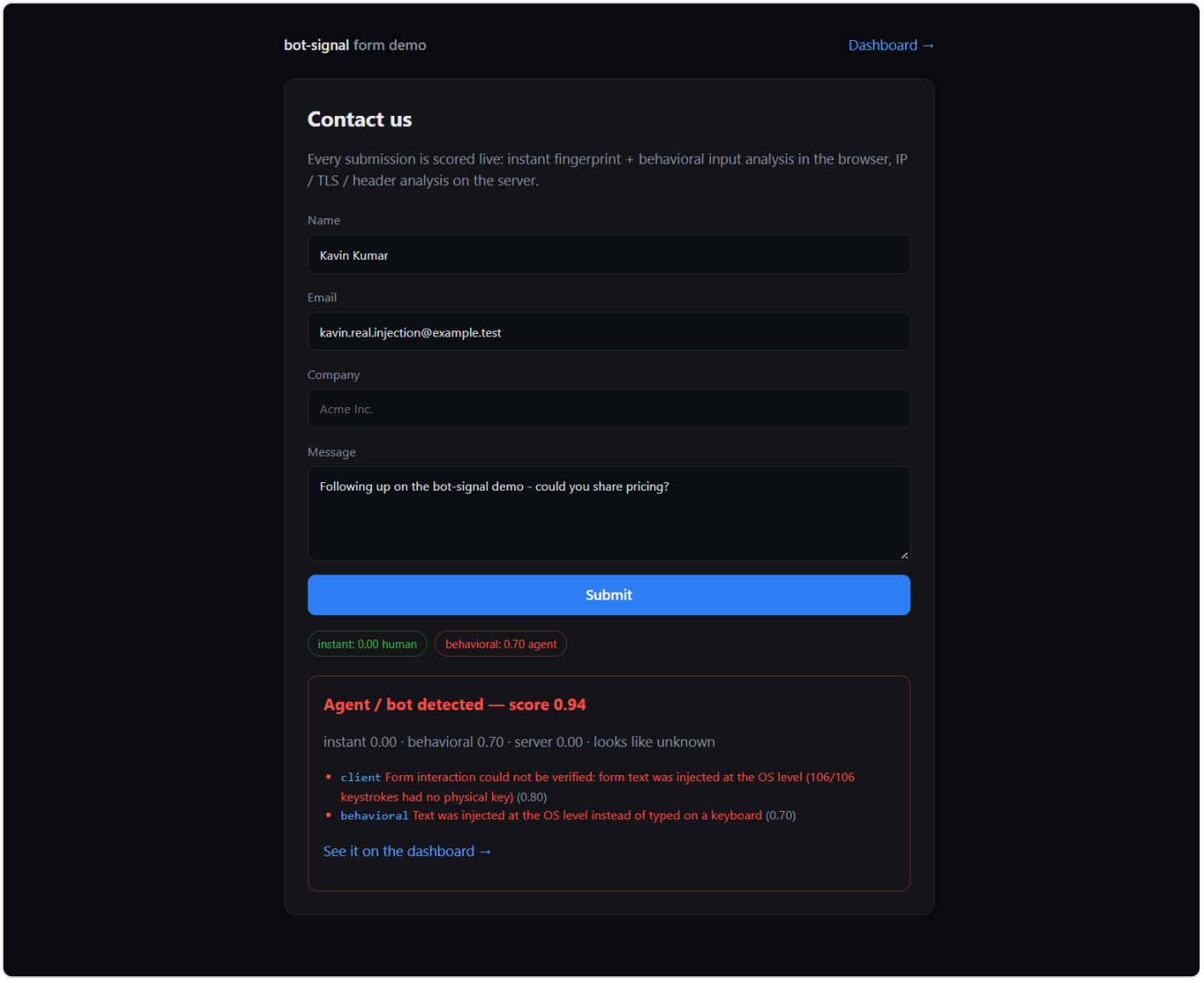
9. OS-level input injection

The sharpest class of agent is one that drives a real, signed-in browser through the Windows input stack — `SendInput`, `keybd_event`, `mouse_event`, `SetCursorPos`, as used by PowerShell and AutoHotkey harnesses. Those events are genuinely `isTrusted`: the browser really did receive them from the operating system, the window is real, the page load is real, and every provenance check passes. Before this run, such a session was classified HUMAN with score 0.00.

The distinguishing artifact is how the text is pushed in. `SendInput` with `KEYEVENTF_UNICODE` sets `wVk = 0` and carries the character in the scan-code field, so Windows reports `VK_PACKET` and Chromium has no physical key to name. Measured directly against Chrome 140 on Windows 11:

INPUT METHOD	KEY	CODE	KEYCODE	ISTRUSTED
`SendInput` + `KEYEVENTF_UNICODE`	K	*(empty)*	231 (`VK_PACKET`)	true
Scan-code key press (hardware path)	b	`KeyB`	66	true

The `injected-key-input` signal fires when at least five printable keystrokes, and at least 60% of them, arrive without a physical key behind them — a ratio, so that an emoji picker or a dictated word inside an otherwise hand-typed form is not flagged. The demo server enforces the same rule on the submitted interaction record, so a tampered page bundle cannot simply drop the evidence.



Real PowerShell `SendInput` session — curved `SetCursorPos` pointer paths, randomised per-character delays, `mouse_event` clicks, genuine page load: now classified AGENT, score 0.94, HTTP 403, on `behavioral:injected-key-input` and `client:untrusted-form-interaction`.

A rewrite that injects real virtual-key/scan-code pairs instead of Unicode packets defeats this specific signal — the keystrokes become indistinguishable from hardware. The supporting `zero-jitter-clicks` heuristic covers part of the pointer side, but this class of attack is not closed by client-side signals alone.

10. Independent red-team results

A separate red-team pass built ten bypass methods against this build and read every verdict back from `GET /api/submissions` rather than the screen. Six passed as human. The findings below are reproduced as stated, with what was changed in response.

FINDING	STATUS	RESPONSE
Client scores are advisory. The server accepted any self-consistent `{score: 0, signals: []}`, so a full page-session replay (no browser at all) and a `Runtime.evaluate` forge from inside a real page both scored human 0.00.	Fixed	The page now submits the raw sample streams behind its verdict, and the server recomputes the behavioral result itself — the reported score no longer decides anything. Three new gates: `missing-behavioral-samples`, `forged-client-verdict` (reported score or signal list disagrees with the samples it was derived from) and `impossible-observation-window` (claimed observation longer than the page session has existed — a duration comparison, so a visitor with a wrong clock is unaffected). Verified against the red team's own scripts: both now score agent 0.99.
The CDP console probe never fires for an attached `Input`-domain client, even with `Runtime.enable`.	Confirmed, open	Reproduced. Three candidate discriminators were measured and *all three rejected*: (1) CDP input skipping Chromium's rAF-aligned pipeline — it does not, 16.9 ms median gap versus 16.8 ms for real OS input, zero sub-frame gaps in either; (2) fewer coalesced raw events than a polled mouse — 1.01 per `pointermove` for both injected paths, and any cadence rule is trivially matched once known; (3) CDP driving an unfocused window — `document.hasFocus()` stays `true`. No discriminator was shipped, because none survived measurement.
`injected-key-input` **is effectively a scan-code check** — it catches `KEYEVENTF_UNICODE` and stock pyautogui, but not `KEYEVENTF_SCANCODE`, CDP `Input`, or USB-HID.	Accurate, open	Correct as stated. Injection that supplies a real scan code is indistinguishable from hardware at the DOM level; there is nothing left in the key event to compare.
`zero-jitter-clicks` **skipped presses held under 40 ms**, so a fast synthetic clicker slipped past regardless of jitter.	Fixed	A press now reads as machine-made if it releases on the pixel it pressed *or* is gone before a finger could lift. Skipping the quick ones only told a clicker to hurry.

What remains undetectable. Four of the six passing methods are still classified HUMAN, and this is architectural rather than an unfinished heuristic: OS-level injection with real scan codes (PowerShell `SendInput`, patched pyautogui) and a raw CDP `Input` client driving a normal Chrome build both produce genuinely trusted events, a real page load and human-grade input. The samples the server now recomputes are honest samples of real input. There is no client-side signal left to add.

Volume and repetition controls

What those vectors cannot hide is repetition — a hand fills this form once, a loop fills it all afternoon. The server now tracks that as a second axis, kept deliberately out of the verdict: `verdict` remains a statement about one submission, `outcome` is what to do about it. A clean-looking submission from an address that has already sent the same content, or too many submissions, or opened too many page sessions, returns `outcome: "review"` and HTTP 202 instead of being silently accepted. Limits are environment-tunable (`MAX_SUBMISSIONS_PER_SESSION`, `MAX_SUBMISSIONS_PER_IP`, `MAX_SESSIONS_PER_IP`, `MAX_CHALLENGES_PER_SESSION`, `MAX_IDENTICAL_SUBMISSIONS`, `RISK_WINDOW_MS`).

Verified against the scan-code injection that defeats detection outright: the first run scored `human` / `accepted`, and the second run with the same content scored `human` / `review`. The verdict stayed honest; the submission stopped being accepted in silence. This is mitigation, not detection, and is reported as such.

A bug this suite could not see

Routing the recomputed behavioral layer through the existing client-layer validation made `invalid-client-layer` fire on *every* samples-backed submission, including honest ones — the validator requires `triggered === true` on each signal, and a full analysis carries untriggered entries too. All 81 tests stayed green, because every case in this matrix expects AGENT, so a verdict that was right for the wrong reason was indistinguishable from one that was right. It surfaced only when a real scan-code injection run came back AGENT on a signal it had no business tripping. A false-positive guard now asserts that a genuine page session carrying real samples trips none of the four sample-integrity gates.

11. Scope note

This is a risk-based set of 75 classification scenarios covering the app's documented layers and the requested Chrome/DOM/JavaScript/request techniques. “All possible” browser and network combinations are not finite; the report does not certify every OS, browser version, mobile device, assistive technology, extension, proxy or TLS terminator.

Chrome DevTools MCP was used directly for native MCP form filling/clicking and MCP-executed `HTMLElement.click()`, `form.requestSubmit()`, and synthetic submit dispatch. The wider matrix additionally used installed Google Chrome through Patchright and raw requests.

17 additional functionality, false-positive-guard and security checks also passed (page/dashboard loading, validation, HTTP handling, storage limits, Unicode, dashboard consistency/reset and output encoding). They are excluded from the headline numbers because they do not answer HUMAN versus AGENT.